

Reporting Summary

Nature Research wishes to improve the reproducibility of the work that we publish. This form provides structure for consistency and transparency in reporting. For further information on Nature Research policies, see [Authors & Referees](#) and the [Editorial Policy Checklist](#).

Statistics

For all statistical analyses, confirm that the following items are present in the figure legend, table legend, main text, or Methods section.

n/a Confirmed

- ☐ ☒ The exact sample size (n) for each experimental group/condition, given as a discrete number and unit of measurement
- ☐ ☒ A statement on whether measurements were taken from distinct samples or whether the same sample was measured repeatedly
- ☐ ☒ The statistical test(s) used AND whether they are one- or two-sided
Only common tests should be described solely by name; describe more complex techniques in the Methods section.
- ☐ ☒ A description of all covariates tested
- ☐ ☒ A description of any assumptions or corrections, such as tests of normality and adjustment for multiple comparisons
- ☐ ☒ A full description of the statistical parameters including central tendency (e.g. means) or other basic estimates (e.g. regression coefficient) AND variation (e.g. standard deviation) or associated estimates of uncertainty (e.g. confidence intervals)
- ☐ ☒ For null hypothesis testing, the test statistic (e.g. F , t , r) with confidence intervals, effect sizes, degrees of freedom and P value noted
Give P values as exact values whenever suitable.
- ☒ ☐ For Bayesian analysis, information on the choice of priors and Markov chain Monte Carlo settings
- ☒ ☐ For hierarchical and complex designs, identification of the appropriate level for tests and full reporting of outcomes
- ☐ ☒ Estimates of effect sizes (e.g. Cohen's d , Pearson's r), indicating how they were calculated

Our web collection on [statistics for biologists](#) contains articles on many of the points above.

Software and code

Policy information about [availability of computer code](#)

Data collection

Behavioral and visual stimulus data were collected using Axoscope 10.6 (Axon Instruments) or WinEDR v3.2.7 (Strathclyde Electrophysiology Software). Two-photon imaging data were collected using PrairieView 5.4 (Bruker).

Data analysis

Two photon imaging data were pre-processed using ImageJ 2.0 (to define regions of interest). All data were analyzed with custom code in python 2.7. Analysis code is available upon request.

For manuscripts utilizing custom algorithms or software that are central to the research but not yet described in published literature, software must be made available to editors/reviewers. We strongly encourage code deposition in a community repository (e.g. GitHub). See the Nature Research [guidelines for submitting code & software](#) for further information.

Data

Policy information about [availability of data](#)

All manuscripts must include a [data availability statement](#). This statement should provide the following information, where applicable:

- Accession codes, unique identifiers, or web links for publicly available datasets
- A list of figures that have associated raw data
- A description of any restrictions on data availability

All data are available upon request.

Field-specific reporting

Please select the one below that is the best fit for your research. If you are not sure, read the appropriate sections before making your selection.

- ☒ Life sciences ☐ Behavioural & social sciences ☐ Ecological, evolutionary & environmental sciences

Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

Sample size	No statistical tests were used to determine sample size. We used sample sizes (~10-20 flies per condition) that have had sufficient statistical power in the past.
Data exclusions	We did not exclude flies or data from any analysis, unless flies appeared unhealthy at the time of the experiment.
Replication	Preliminary experiments produced the same findings as those described in the paper.
Randomization	Flies were randomly chosen for each experiment.
Blinding	The investigators were not blind to the flies' genotypes. All data collection and analysis was done computationally.

Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

Materials & experimental systems		Methods	
n/a	Involved in the study	n/a	Involved in the study
☐	☒ Antibodies	☒	☐ ChIP-seq
☒	☐ Eukaryotic cell lines	☒	☐ Flow cytometry
☒	☐ Palaeontology	☒	☐ MRI-based neuroimaging
☐	☒ Animals and other organisms		
☒	☐ Human research participants		
☒	☐ Clinical data		

Antibodies

Antibodies used	anti-Brp (1:30, DSHB nc82), rabbit anti-HA Tag (1:300, Cell Signaling #3724S), Rat anti-FLAG Tag (1:200, Novus #NBP1-06712), Dylight550 mouse anti-V5 Tag (1:500, AbD Serotec MCA1360D550GA), AlexaFluor488 goat anti-mouse (1:400, Life Technologies #A11029), AlexaFluor594 donkey anti-rabbit (1:500, Jackson Immuno Research # 711-585-152) , AlexaFluor647 donkey anti-rat (1:150, Jackson Immuno Research #712-605-153)
Validation	The antibodies were used to detect transgenically-expressed proteins for multicolor flip-out experiments. These antibodies labeled different cell types if these transgenes were expressed from different Gal4 lines, demonstrating that the antibodies were binding as expected, and not non-specifically.

Animals and other organisms

Policy information about [studies involving animals](#); [ARRIVE guidelines](#) recommended for reporting animal research

Laboratory animals	We used Drosophila melanogaster. All flies were female. Fly genotypes were: Canton-S, from Martin Heisenberg, through Michael Dickinson pJFRC99-20XUAS-IVS-Syn21-Shibire-ts1-p10 inserted at VK00005, from Gerald Rubin UAS- P2X2, from Vanessa Ruta LexAOp-GCaMP6f, BDSC #44277 UAS-GCaMP6f, BDSC #52869 VT032906-Gal4, VDRC #202537 60D05-LexA, BDSC #52867 60D05-Gal4, BDSC #39247 77E05-Gal4, BDSC #48338 27F02-Gal4, BDSC #48072 57C10-FLPL / + ; + ; 10xUAS-FRT.stop-myr::smGdP-HA, UAS-FRT.stop-myr::smGdP-V5-THS-10xUAS-FRT.stop-myr::smGdP-FLAG , BDSC #64087 57C10-FLPG5 / + ; + ; 10xUAS-FRT.stop-myr::smGdP-HA, UAS-FRT.stop-myr::smGdP-V5-THS-10xUAS-FRT.stop-myr::smGdP-FLAG, BDSC #64088
Wild animals	The study did not involve wild animals.

Field-collected samples

The study did not involve samples collected from the field.

Ethics oversight

No ethical approval is required for work in *Drosophila melanogaster*.

Note that full information on the approval of the study protocol must also be provided in the manuscript.